Yixiao Chi

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Education

MS **Carnegie Mellon University**, Information Networking — Aug 2026 – present

BS **Zhejiang University**, Computer Science and Technology — Sept 2022 – June 2026

- GPA: 4.02/4.3, Major GPA: 4.14/4.3

Experience

VIPA Lab, Zhejiang University, Research Intern — Dec 2023 – Aug 2025

MVIG-RHOS Lab, Shanghai Jiao Tong University, Research Intern — Aug 2025 – Nov 2025

ACE, Sensetime, Algorithm Engineer Intern — Mar 2026 – June 2026

Projects

Preserving history with intelligence — July 2023 – June 2024

- Advisor: Wei Chen, Professor in State Key Lab of CAD&CG, Zhejiang University.
- Accomplishments:
- China International College Students' Innovation Competition (2024) *Bronze Prize*
- **Granted Chinese Patent**: "Deep Learning-Based Archaeological Information Extraction Method, Apparatus, Equipment, and Medium" (CN119693633B) Issued April 29, 2025.
- Developed a deep learning pipeline integrating YOLO object detection and PaddleOCR for automated extraction, segmentation, and semantic naming of archaeological figures and annotations from document images, significantly enhancing data processing efficiency and implemented an interactive image segmentation module based on the SAM model to enable precise manual refinement of extraction results.

MiniSQL — Apr 2024 – June 2024

A course project where we developed a database engine based on the CMU 15-445 BusTub framework. I implemented the B+ Tree Index Manager, Lock Manager and Transaction Manager.

- B+ Tree Index Manager: Developed comprehensive tree functionality, including $O(\log n)$ search/query operations (point queries and range scans) and crucial maintenance algorithms. These mechanisms cover full node splitting (Split) on insertion and complex coalesce (merge) and key redistribution during deletion.
- Lock Manager: Engineered a full Two-Phase Locking (2PL) protocol implementation, handling requests for Shared, Exclusive, and Upgrade locks using per-data-item lock request queues. Developed a Deadlock Detection thread that periodically built and analyzed the Waits-For Graph using DFS to detect and resolve cycles via aborting the youngest transaction, ensuring system liveness.
- Transaction Manager: Implemented the core Transaction Manager responsible for concurrency orchestration, including the creation, lifecycle tracking, commitment, and abortion of transactions under various Isolation Levels.

Snake 3v3 — Aug 2024 – Sept 2024

Achieved top performance in class competition: The Heuristic Algorithm (Rule-based policy) **ranked 1st**, and the QMIX reinforcement learning policy **ranked 2nd**.

- Developed and implemented QMIX and MAPPO strategies for a 3v3 Snake game.
- Designed and tuned a complex Reward Shaping scheme, including specific rewards for eating food, penalties for opponents, and adjustments based on snake length differences.
- Applied Population Based Training (PBT) for dynamic hyperparameter tuning and exploring the strategy space during training, leveraging parallel data collection.

Price Compare Website Oct 2024 – Jan 2025

Independently designed and developed a full-stack web application for product price comparison across major e-commerce platforms (Tianmao, JD, Pinduoduo).

- Implemented core features including user authentication, personal product libraries, historical price tracking and charting, and automated price drop notifications.
- Developed a data acquisition module using Selenium to collect real-time pricing from online users.

Video quality evaluation of graphic rendering based on deep learning Nov 2024 – May 2025

- Advisor: Xiaogang Jin, Professor in State Key Lab of CAD&CG, Zhejiang University.
- Constructed the **first large-scale dataset** (*ReVQ-2k*) for rendered video quality (2,000 videos, 57k subjective scores), uniquely including temporal stability annotations crucial for assessing rendered content.
- Co-developed a novel two-stream *NR-VQA* model combining image quality assessment (Swin-T based) with a dedicated temporal stability analysis module (utilizing motion estimation and image differencing), achieving state-of-the-art performance for rendered videos.

Groove Engine Aug 2025 – Sept 2025

Douyin Hackathon 2025 Finalist (Top 24 Nationwide)

- Built an AI-based music therapy application, aiming to reduce user anxiety by generating personalized music playlists from heart rate variability (HRV) data.
- Established a full-stack web application using Vite + React for frontend and Django for backend, designed efficient RESTful APIS and deployed the service on an Aliyun server **within 48 hours**.
- Engineered a hybrid music recommendation pipeline that converted 28 HRV physiological features into VA emotional coordinates for initial song similarity matching, followed by integrating a Doubao Large Model for user-type-driven secondary personalization.

Publications

BiTrajDiff: Bidirectional Trajectory Generation with Diffusion Models for Offline Reinforcement Learning July 2026

Yunpeng Qing[1*], **Yixiao Chi**[2*], Shuo Chen[2*], Shunyu Liu[3], Sixu Lin[4], Changqing Zou[1,5†]

arxiv.org/abs/2506.05762 (ICML 2026)

ACE-Brain-0.5: A Unified Embodied Foundational Model for Physical Agentic AI July 2026

ACE-Brain Team(**Yixiao Chi** is a core contributor)

arxiv.org/abs/2607.04426 (Technical Report)

Selected Awards

- Third-class Scholarship (2023, 2024, 2025)
- The Mathematical Contest in Modeling (MCM 2024) Meritorious Winner
- 2023 Zhejiang Provincial Higher Mathematics Competition for University Students(Engineering) Second Prize
- 15th National College Student Mathematics Competition (Non-Mathematics Category) Second Prize

Skills

Languages: Python, C++, C

ML Frameworks: PyTorch, JAX

Research Areas: Reinforcement Learning, Vision-Language-Action Models, Embodied Intelligence, Robotics